

1. Administrative & Study Identification

Full Study Title: A Phase III Study to Investigate Efficacy, Safety and Tolerability of Iptacopan Compared With Placebo in Participants Aged 18 to 75 Years With gMG. **Short Title/Acronym:** Iptacopan in gMG (CLNP023Q12301) **Protocol Number:** CLNP023Q12301 / 2023-507064-39-00 **NCT Number:** NCT06517758 **Trial Status:** RECRUITING **Sponsor/Company Name:** Novartis **Investigational Product:** Iptacopan hydrochloride (oral, proximal alternative complement inhibitor targeting Factor B) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of iptacopan compared to placebo in participants with anti-acetylcholine receptor antibody-positive (AChR+) generalized Myasthenia Gravis (gMG) on stable standard of care (SOC) treatment, measured by the change in the Myasthenia Gravis Activities of Daily Living (MG-ADL) score. **Secondary Objectives:** To evaluate the safety, tolerability, and effect of iptacopan on secondary clinical measures including Quantitative Myasthenia Gravis (QMG) score and responder rates.

2.2 Background & Rationale To address the need for targeted therapies in patients with gMG who have inadequate disease control on standard non-steroidal immunosuppressants or require frequent rescue therapies (e.g., plasmapheresis, IVIG). Iptacopan blocks a part of the immune system involved in the disease (Factor B in the alternative complement pathway) and has shown promise in improving daily task performance for patients with gMG.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1, Stratified by region

3.2 Planned Enrollment & Duration Target \$N\$: 146 evaluable patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 2024 **Estimated Primary Completion:** 6 months (180 days) for the primary double-blind period, followed by a 24-month open-label extension (OLE) period.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients with generalized Myasthenia Gravis (age 18-75 years). 2. Positive serology testing for AChR+ antibody at screening. 3. Myasthenia Gravis Foundation of America (MGFA) Class II-IV gMG and likely not in need of a respirator for the duration of the study. 4. Confirmation of gMG diagnosis supported by ≥ 1 of the following: history of abnormal neuromuscular transmission (single-fiber EMG or repetitive nerve stimulation), history of positive edrophonium chloride test, or demonstrated improvement in MG signs on oral acetylcholinesterase inhibitors. 5. Baseline MG-ADL score ≥ 6 , with $\geq 50\%$ of the total score due to non-ocular symptoms. 6. Not optimally controlled for ≥ 6 months on just one NSIST; or two or more NSISTs; or on frequent (at least quarterly) plasmapheresis, plasma exchange, or IVIG; or on an approved gMG treatment (FcRn antagonist, rituximab, or other therapies excluding complement inhibitors). 7. Vaccinated against *Neisseria meningitidis* and *Streptococcus pneumoniae*. Also vaccinated against *Haemophilus influenzae* depending on local regulations. Vaccinations must be performed at least 2 weeks prior to first dosing, or prophylactic antibiotic treatment must be initiated until 2 weeks post-vaccination.

4.2 Exclusion Criteria 1. Treatment with IVIG/PLEX in the past month, rituximab in the past 6 months, eculizumab in the past 2 months, ravulizumab/other complement inhibitors in the past 3 months, efgartigimod/anti-FcRn therapies in the past 3 months, or thymectomy in the past 6 months (or planned during the trial). 2. Clinically significant active or chronic uncontrolled bacterial, viral, or fungal infection at screening (including Active HBV, Active HCV, or HIV positive with AIDS-defining condition or CD4 count < 200 cells/mm³). 3. Active systemic bacterial, viral (including COVID-19), or fungal infection, or any major episode of infection requiring hospitalization or injectable antimicrobial therapy within 14 days prior to study drug administration. 4. Presence of fever $\geq 38^\circ\text{C}$ (100.4°F) within 7 days prior to study drug administration. 5. History of recurrent invasive infections caused by encapsulated organisms (e.g., *N. meningitidis* and *S. pneumoniae*). 6. Female participants who are pregnant, lactating, or intending to become pregnant; and women of child-bearing potential not using effective methods of contraception.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iptacopan hydrochloride or matching placebo taken orally. **Route of Administration:** Oral **Duration of Treatment:** 6 months (180 days) in the double-blind phase, followed by an open-label extension phase of 24 months.

5.2 Concomitant & Prohibited Therapy Permitted: Stable standard of care (SOC) treatments for gMG. **Prohibited:** Initiation of new immunosuppressants or unauthorized rescue therapies during the 6-month double-blind period without investigator approval.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Infection/Vaccination verification
Baseline	Day 0	Randomization, First Dose, Baseline MG-ADL/QMG assessments
Interim	Over 6 months	Safety Labs, Efficacy Assessments (MG-ADL, QMG)
End of Study	Month 6	Primary Endpoint evaluation; entry into OLE or Safety Follow-up (7 and 30 days post-treatment)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline to Month 6 in Myasthenia Gravis Activity of Daily Living (MG-ADL) total score. **Measurement Method:** The MG-ADL is an 8-item interviewer-led patient reporting scale assessing functional disability secondary to ocular, bulbar, respiratory, and gross motor/limb impairment. Each item is scored 0-3. The total score ranges from 0 to 24, with higher scores indicating more disability.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline to Month 6 in Quantitative MG (QMG) total score. 2. Proportion of participants with ≥ 5 points reduction from baseline to Month 6 of QMG total score without rescue medication and strongly confounding prohibited medication. 3. Proportion of participants with ≥ 3 points reduction from baseline to Month 6 of MG-ADL total score without rescue medication and strongly confounding prohibited medication. 4. Overall safety and tolerability profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via scheduled visits, with special attention to signs of encapsulated bacterial infections. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Independent safety review committee to monitor infection risks and overall trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE evaluations. **Sample Size Justification:** $N=146$ to achieve sufficient power ($\alpha = 0.05$) to detect a statistically significant change in the MG-ADL score between iptacopan and placebo groups. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) or Multiple Imputation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to assure protocol adherence. **Audit Trail:** Complete eCRF locking and query resolution via data clarification forms.

11. Study Identifiers & Governance

Primary ID: CLNP023Q12301 **Registry Number:** NCT06517758 **Posting ID:** 1102717460527349771 **Sponsor/Lead Organization:** Novartis **Study Title:** A Phase III Study to Investigate Efficacy, Safety and Tolerability of Iptacopan Compared With Placebo in Participants Aged 18 to 75 Years With gMG. **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Phase III Study to Evaluate the Efficacy, Safety, and Tolerability of Iptacopan in Patients With Generalized Myasthenia Gravis (gMG), Followed by an Open Label Extension Phase.

12. Clinical Framework (gMG Specific)

Disease Area: Generalized Myasthenia Gravis **Primary Condition:** Generalized Myasthenia Gravis (gMG) **Diagnosis Threshold:** Anti-AChR+ status, MGFA Class II-IV, Baseline MG-ADL ≥ 6 **Medical Methodology:** Alternative complement pathway inhibition (Factor B) combined with stable standard of care (SOC) **Optimization Target:** Significant reduction in disease severity and improvement in daily living activities (MG-ADL)

13. Study Procedures & Methodology

Management Pathway: Oral complement inhibitor administration alongside existing stable SOC **Education Protocol (HCP):** Mandatory training on proper administration of MG-ADL and QMG clinical assessments **Education Protocol (Patient):** Pre-enrollment counseling regarding mandatory vaccinations (*N. meningitidis*, *S. pneumoniae*, *H. influenzae*) and infection signs **Quality Audit**

Mechanism: Site initiation visits, regular compliance checks, and drug accountability logs

14. Observation & Follow-Up Schedule

Total Duration: 6 months (double-blind core) + 24 months (Open-Label Extension) **Onsite Visit Cadence:** Screening, Baseline, regular intervals up to Month 6, followed by safety assessments at 7 and 30 days post-treatment **Remote Monitoring Frequency:** Every 4 weeks (Q4W) **Checklist Review Interval:** Every 12 weeks (Q12W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				